Business Intelligence Systems

INFO 4555 / 5556

Fall 2025

Order Management

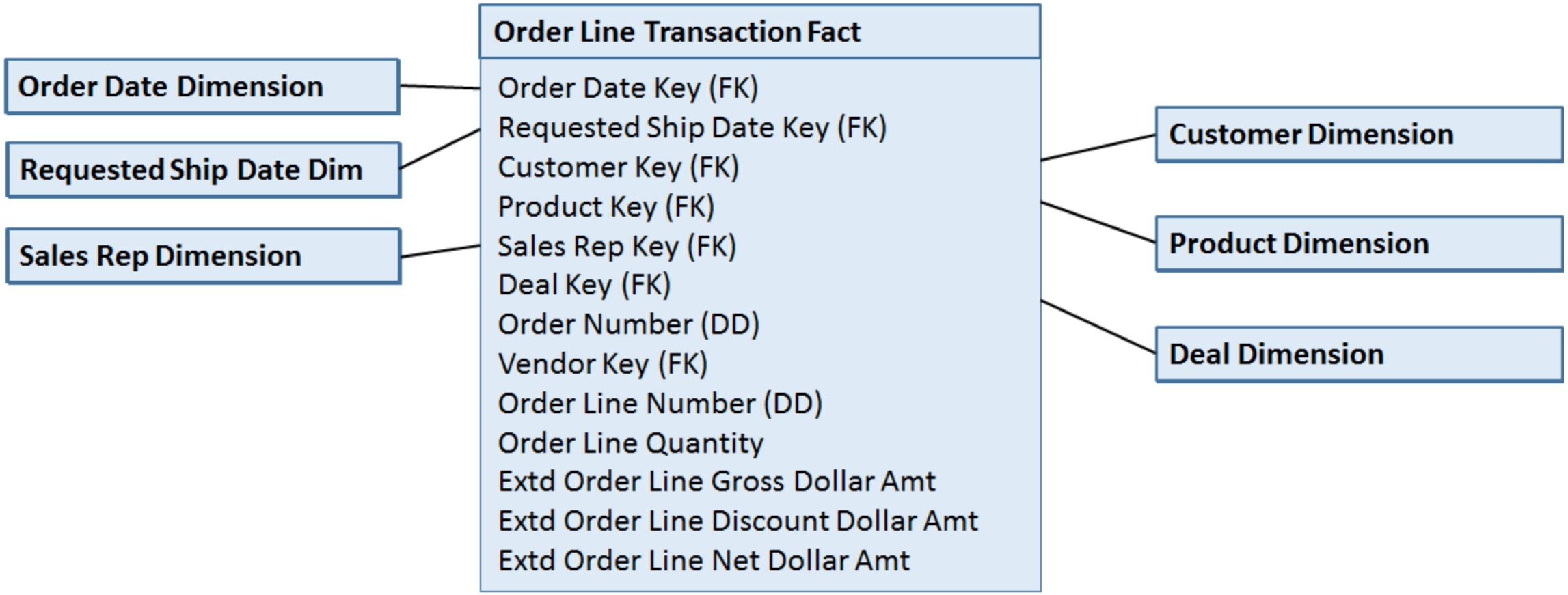


Order Management

Order management often consists of the following business processes:

- Quoting
- Ordering
- Shipping to customer
- Shipment invoicing
- Receiving payments
- Customer returns

Kimball Order Line Transaction Fact



The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 6-2

Fact Table Normalization

- Rather than having multiple facts in the same fact table, a “transaction type” dimension is added and a single generic fact attribute is used to represent the measure.
- Normalization of a fact table is generally not advised because:
  - Increases the number of records in the fact table Times the number of fact types
  - Adds complexity to queries against the normalized fact table
- The one scenario where Kimball mentions normalization of the fact might be appropriate is when the facts are sparsely populated within the rows and analysis of the fact types is independent of one another.

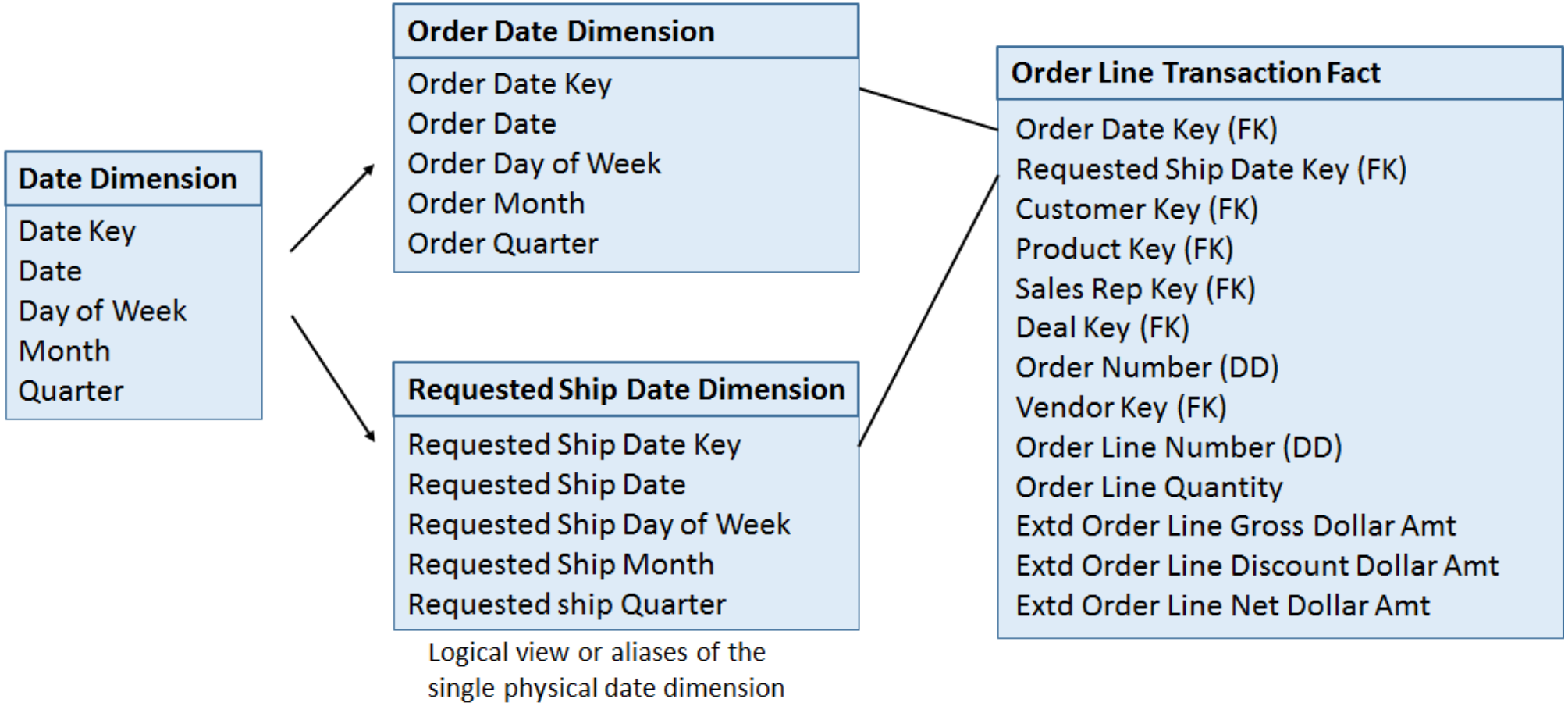


Dimension Role Playing

- Revisited from chapter 3.
- Often there will be multiple date references in a fact table
- SQL does not easily allow the joining of multiple dates in a fact to a single date table
- Role play is the creation of database views to assign alternate names to the date dimension table and its attributes.
- Role play allows the use of one physical table multiple times within a single query
- Simplifies the query for user and BI tools



Kimball Sample Dimension Role Play



The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 6-3

Communicating role-playing dimensions on the Bus Matrix

	Date
Quoting	Quote Date
Ordering	Order Date Requested Ship Date
Shipping to Customer	Shipment Date
Shipment Invoicing	Invoice Date
Receiving Payments	Payment Receipt Date
Customer Returns	Return Date

The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 6-4

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All of the following statements are true about Dimension Role Playing, except:

A: Allows the user to query one physical table multiple times within a single query

B: Simplifies the queries for user and BI tools

C: Results in significant query response time gains

D: Often used to represent multiple references to date in a fact table

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Steps to creating the Product dimension off an operational product master table

- Remap the operational product code to a surrogate key
- Add descriptive attribute values to augment or replace operational codes
- Quality check the attribute values to ensure no misspellings, impossible values, or multiple variations
- Document the attribute definitions, interpretations, and origins in the metadata

Should Sales Rep attributes be included in the Customer Dimension? (considerations)

- Is the one-to-one or many-to-one relationship actually a many-to-many?
- Does the sales rep and customer relationship vary over time or under the influence of another dimension?
- Is the customer dimension extremely large?
- Do the sales rep and customer dimensions participate independently in other fact tables?
- Does the business think about the sales rep and customer as separate things?

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Do you feel comfortable with the discussion about how to determine whether to break the Sales Rep attributes into a separate dimension?

Yes – Lets keep moving through the material

Maybe – Good enough, keep going

NO – Need more explanation

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How to handle miscellaneous indicators and flags that have a small range of discrete values?
(a few options that are not recommended)

- Ignore the flags and indicators
- Leave the flags and indicators unchanged on the fact row
- Make each flag and indicator into its own dimension
- Store the flags and indicators in an order header dimension

Junk Dimensions



- similar to the junk drawer in the kitchen, it is a place to store some of the needed attributes that are not appropriately fit with other dimensions, yet don’t justify their own individual dimensions.
- contains a set of attributes that are not necessarily related.
- contains all of the possible combinations of the attribute values
- avoid too many attributes in a single junk dimension. The number of possible combinations can become extremely large.

Sample Junk Dimension

Order Indicator Key	Payment Type Description	Payment Type Group	Order Type	Commission Credit Indicator
1	Cash	Cash	Inbound	Commissionable
2	Cash	Cash	Inbound	Non-Commissionable
3	Cash	Cash	Outbound	Commissionable
4	Cash	Cash	Outbound	Non-Commissionable
5	Visa	Credit	Inbound	Commissionable
6	Visa	Credit	Inbound	Non-Commissionable
7	Visa	Credit	Outbound	Commissionable
8	Visa	Credit	Outbound	Non-Commissionable
9	MasterCard	Credit	Inbound	Commissionable
10	MasterCard	Credit	Inbound	Non-Commissionable
11	MasterCard	Credit	Outbound	Commissionable
12	MasterCard	Credit	Outbound	Non-Commissionable

The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 6-8

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Do you feel comfortable with the concept of "Junk Dimensions"?

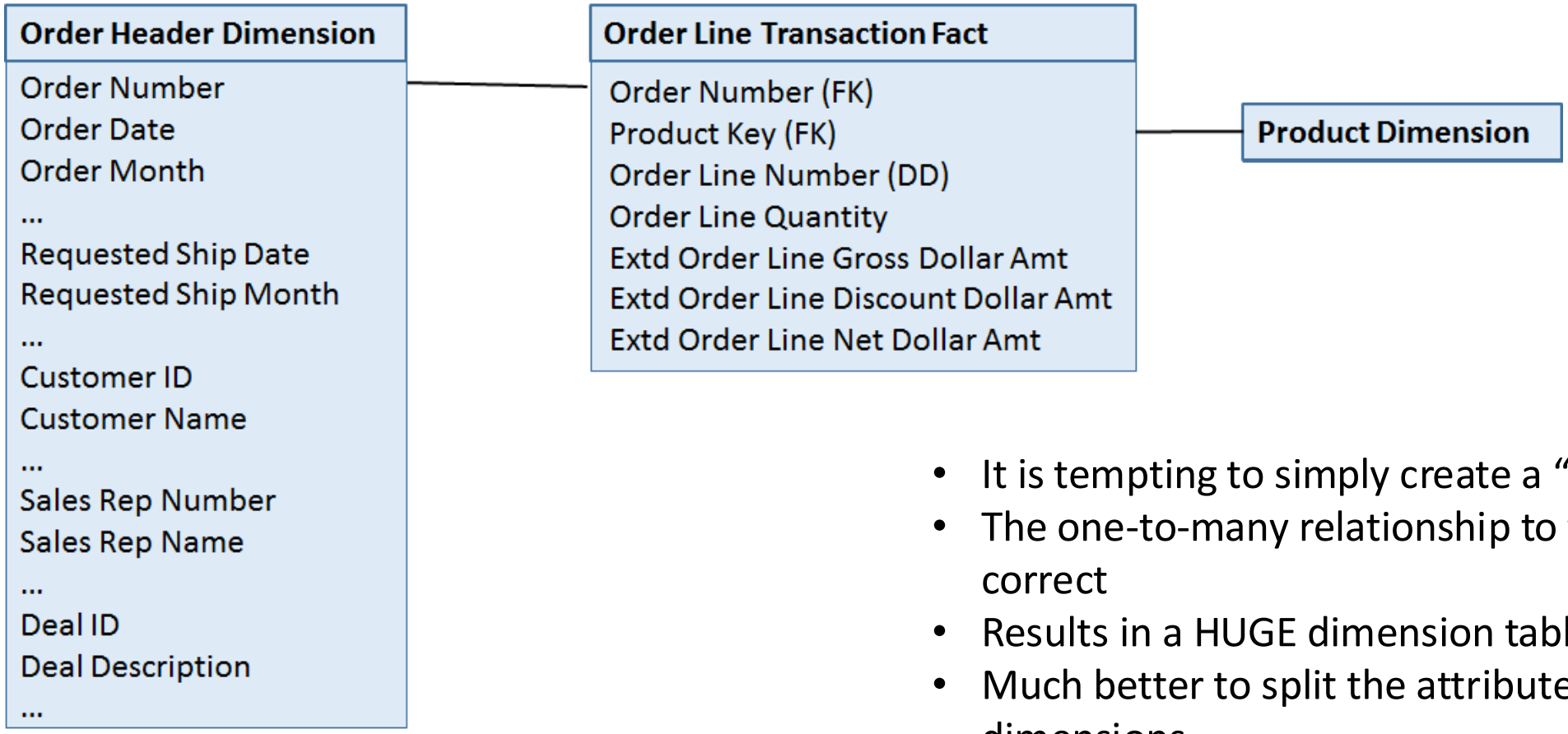
Yes – I now know where to store my calculator and stamps

Maybe – Good enough, still not sure where to store the stamps

NO – Need more explanation, my entire kitchen is a junk drawer

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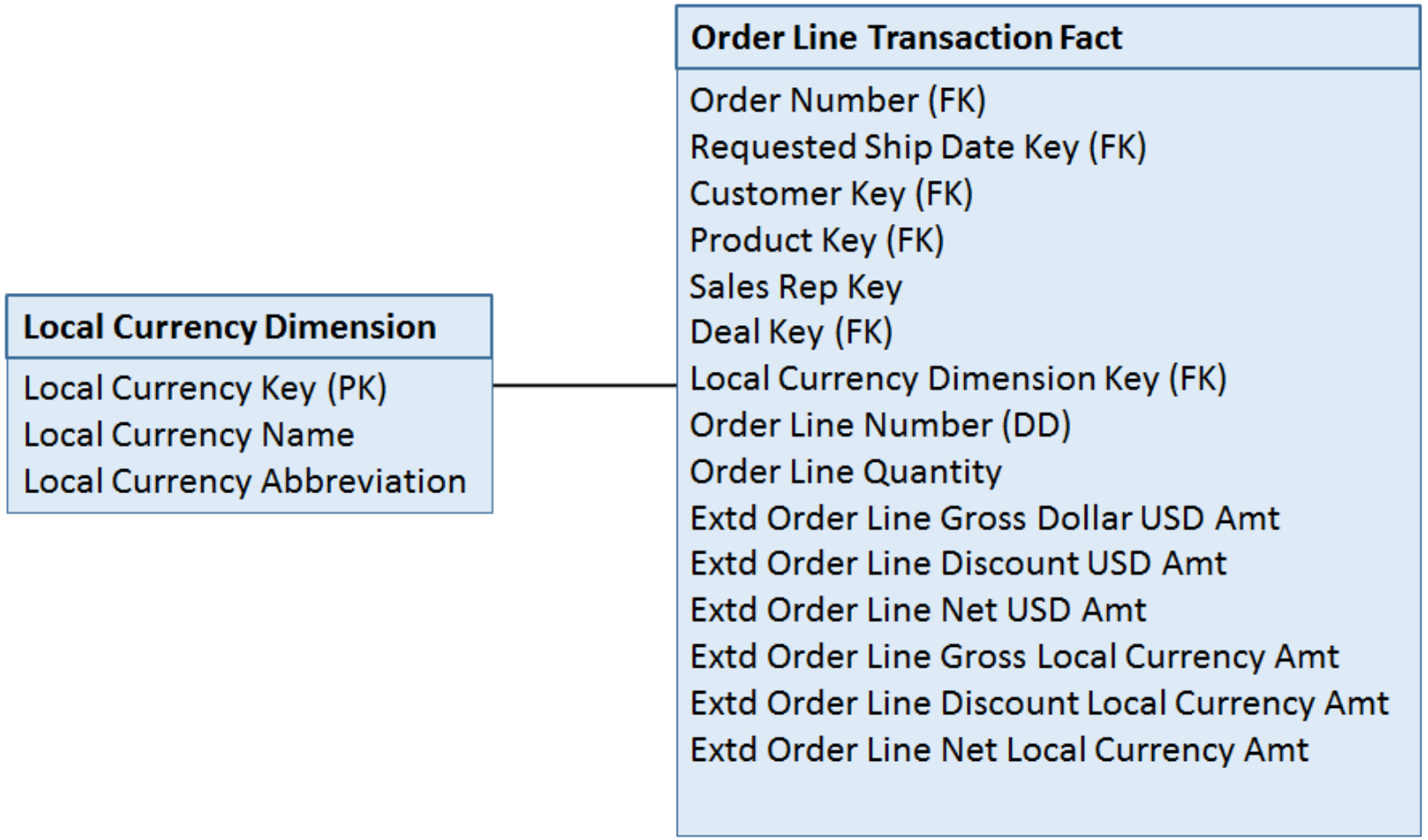
Do NOT create a Header Dimension



- It is tempting to simply create a “header” dimension
- The one-to-many relationship to fact “line item” table is correct
- Results in a HUGE dimension table (many records)
- Much better to split the attributes out into logical dimensions

The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 6-9

Kimball approach to Multiple Currencies



The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 6-10

- Capture transactions in the local currency (actual value)
- Capture the transaction currency values and the standardized corporate currency
- Need to determine rules for conversion rate. Ex. Exchange rate at time of transaction, end of business day, etc.

Use of Currency Conversion Fact

Currency Conversion Fact
Conversion Date Key (FK)
Source Currency Key (FK)
Destination Currency Key (FK)
Source-Destination Exchange Rate
Destination-Source Exchange Rate

Allows the tracking of multiple currency exchange rates on a daily basis

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Do you feel comfortable with the conversation about multiple currencies?

Yes – Lets keep moving through the material

Maybe – Good enough, keep going

NO – Need more explanation

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Transaction Facts at Different Granularity

- Example: an order will likely have multiple line items, but a single shipping charge associated with it. Two grains.
- “Allocation” is the process of breaking these header level costs up and associating them across multiple line items.
- Allocation rules can become very complex and often difficult to get consensus on rules
- Allocation rules are another challenge for data governance.
- Do not mix fact granularities such as order header and order line facts within a single fact table. If allocation is not an option, split into multiple fact tables.

Allocating Header Cost example

If the header cost of "shipping charges" can be allocated across the order line items, the separate Order Header Transaction Fact is not needed.

Order Header Transaction Fact
Order Date Key (FK)
Requested Ship Date Key (FK)
Customer Key (FK)
Product Key (FK)
Sales Rep Key
Deal Key (FK)
Order Number (PK)
Order Shipping Charges Dollar Amt

Allocated

Order Line Transaction Fact
Order Date Key (FK)
Requested Ship Date Key (FK)
Customer Key (FK)
Product Key (FK)
Sales Rep Key
Deal Key (FK)
Order Number (DD)
Order Line Number (DD)
Order Line Quantity
Extd Order Line Gross Dollar Amt
Extd Order Line Discount Dollar Amt
Extd Order Line Net Dollar Amt
Allocated Order Line Shipping Charges Dollar Amount

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Do you feel comfortable with the conversation about Allocating Header Costs?

Yes – Lets keep moving through the material

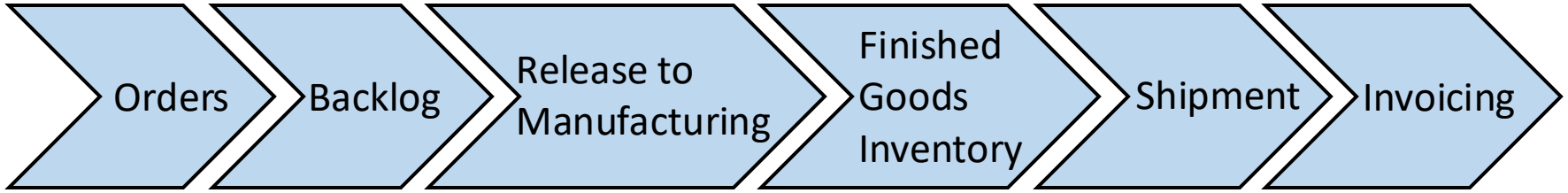
Maybe – Good enough, keep going

NO – Need more explanation

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Accumulating Snapshot for Order Fulfillment

- The Order Fulfillment Pipeline lends itself to measuring the performance of the pipeline via an Accumulating Snapshot
- Sample Accumulating Fact on page 195 fig 6-19



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Accumulating Snapshots and Type 2 Dimensions

- Accumulating snapshots present the latest state of a workflow or pipeline
- If a related dimension attributes being tracked by type 2 is changed before an accumulating fact row has reached a completed state, the associated dimension key should be updated to the current key
- Once a record in an accumulating fact table has reached a completed state it is generally not updated to reflect current dimension value status.

Lag Calculations in Accumulating Fact Tables

- often more complicated than simply subtracting begin date from end date.
- often need to account for weekend days, holidays, etc

Multiple Units of Measure

- Different functional organization may want to see performance metrics in different units of measure. Ex. Pallets, retail cases, scan units, or individual packaging.
- Recommended approach is to identify unit measures (facts) at the smallest unit measure, then provide additional facts for the conversion factors of the other possible measurement types.
- Solution should be rolled out using views over the fact table for the various measure types. (no need for the customers to convert)

Multiple units of Measure (example)

Order Fulfillment Accumulating Fact
Date Keys (FKs)
Product key (FK)
More FKs…
Order Quantity Shipping Cases
Release to Manufacturing Quantity Shipping Cases
Manufacturing Pass Inspection Quantity Shipping Cases
Manufacturing Fail Inspection Quantity Shipping Cases
Finished Goods Inventory Quantity Shipping Cases
Authorized to Sell Quantity Shipping Cases
Shipment Quantity Shipping Cases
Shipment Damage Quantity Shipping Cases
Customer Return Quantity Shipping Cases
Invoice Quantity Shipping Cases
Pallet Conversion Factor
Retail Cases Conversion Factor
Scan Units Conversion Factor
Equivalized Consumer Units Conversion Factor

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Now that we have completed the slides for Chapter 6 Order management, are you comfortable with the material?

YES

OK – with a little review of my notes and slides I will be ready for quiz

NO

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